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INTRODUCTION

The NT7086PQ is a LCD driver LSI that is fabricated by low power CMOS high voltage process technology.

In segment drive mode, it can be interfaced in 1-bit serial or 4-bit parallel method by the controller.

In common drive mode, dual type mode is applicable. And in segment mode application, the power down function reduces power consumption.

FEATURES

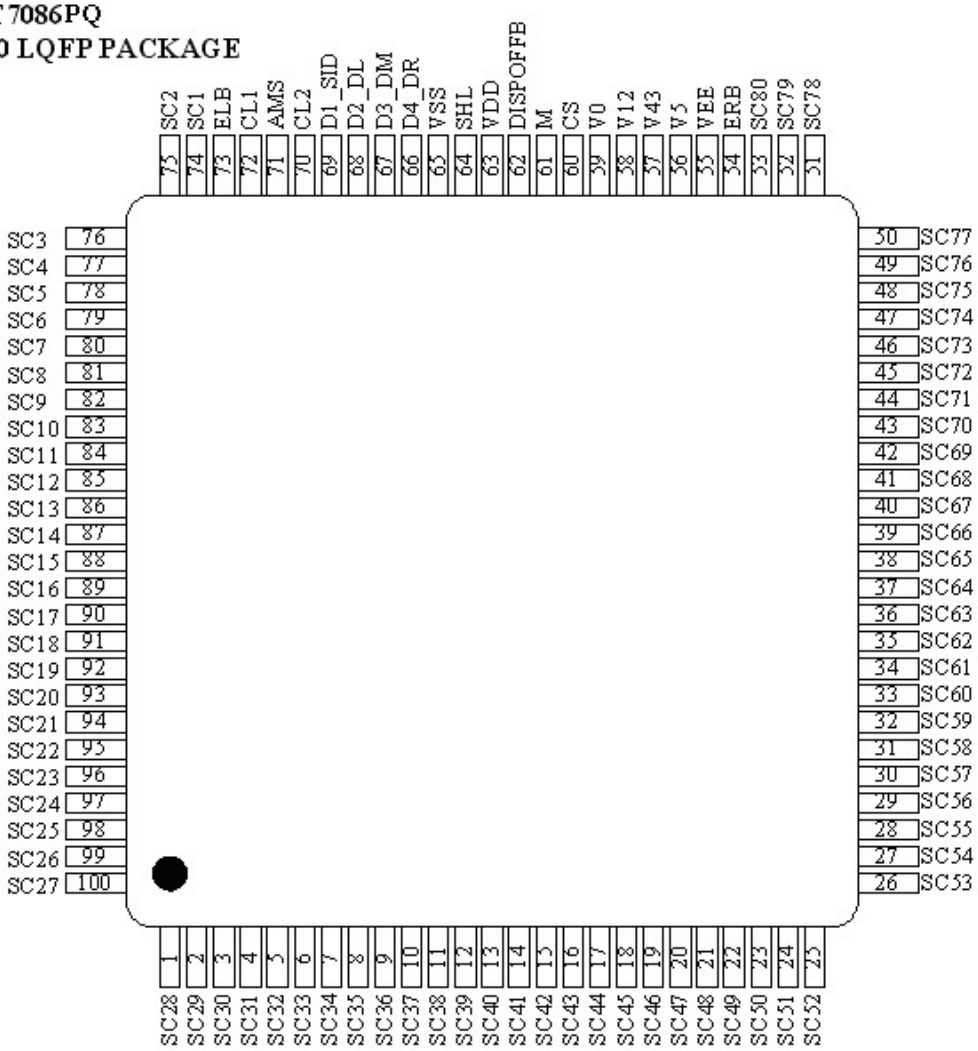
- Power supply voltage:+5 V±10%, + 3V±10%
- Supply voltage for display: 6 to 28 V (V_{DD}-V_{EE})
- 4-bit parallel/1-bit serial data processing (in segment mode)..
- Single mode operation / dual mode operation (in common mode).
- Power down function (in segment mode).
- Applicable LCD duty:1/64 ~ 1/256
- Interface

DRIVERS	
COM(cascade)	**SEG(cascade)**
NT7086PQ	NT7086PQ

- High voltage CMOS process.
- Available PKG type : bare chip, 100-LQFP, 100-QFP

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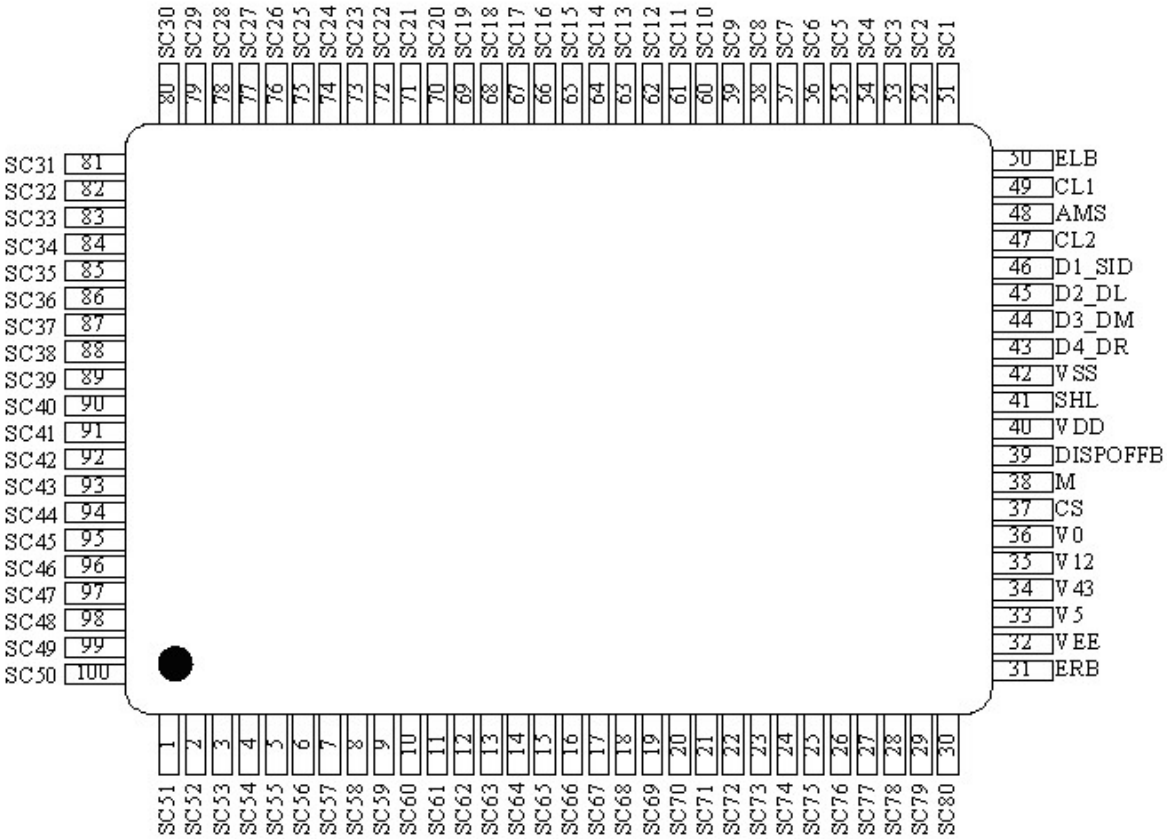
NT7086PQ
100 LQFP PACKAGE



PKG TYPE	=	100-LQFP
PKG THICKNESS	=	1.40 (± 0.05) mm
PKG SIZE	=	14.00 (± 0.10) X 14.00 (± 0.10) mm
PAD PITCH	=	0.5 mm
PAD WIDTH	=	0.20 (+0.07,-0.03) mm
PAD LENGTH	=	1.0 (±0.1) mm

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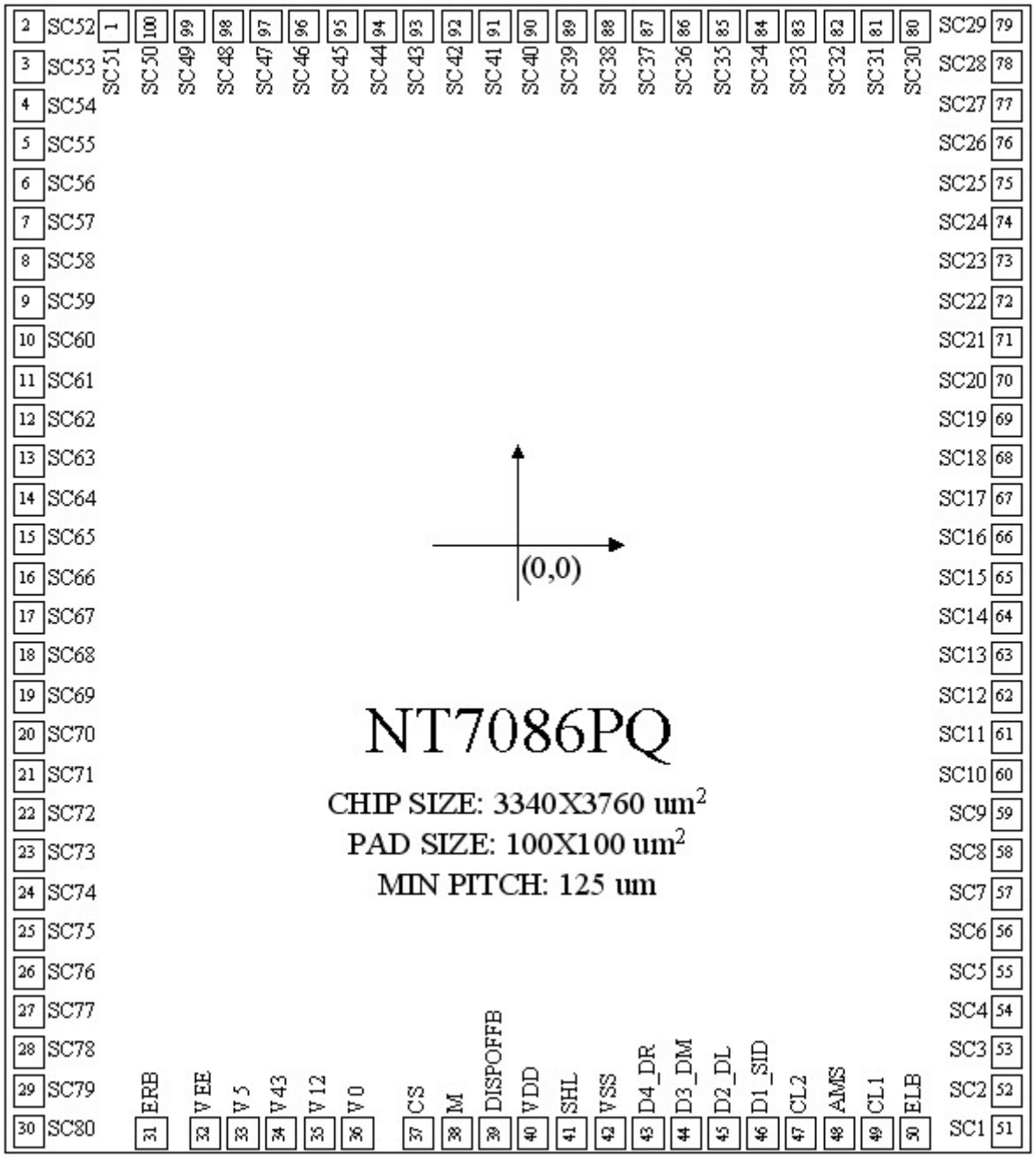
NT7086PQ
100 QFP PACKAGE



PKG TYPE	=	100-QFP
PKG THICKNESS	=	3.00 (± 0.40) mm
PKG SIZE	=	24.80(± 0.20) X 18.8 (± 0.20) mm
PAD PITCH	=	0.65 mm
PAD WIDTH	=	0.30 (± 0.10) mm
PAD LENGTH	=	2.4 (± 0.20) mm

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PAD DIAGRAM ***Note:*** Please connects the substrate to V_{DD} or Floating



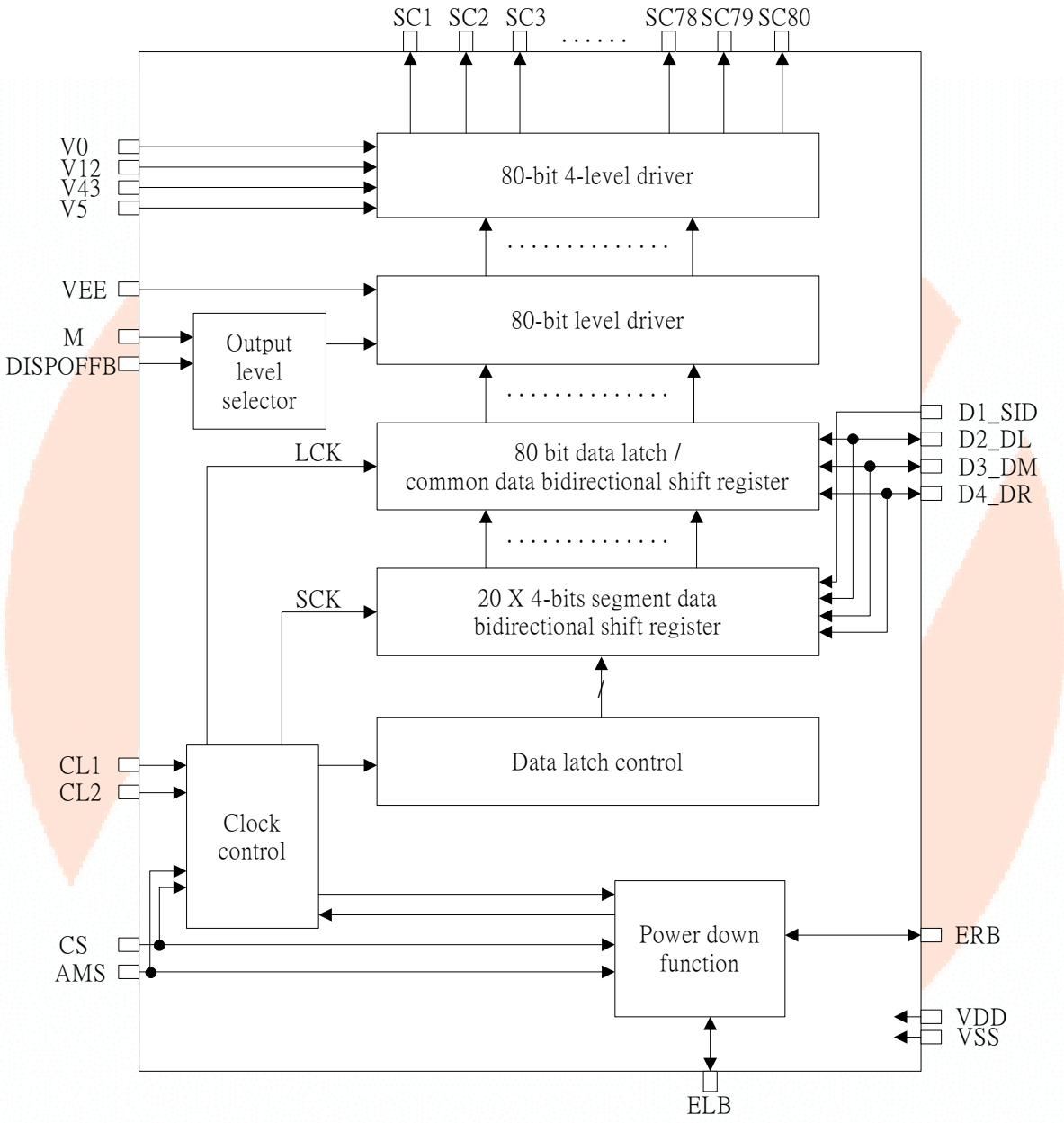
Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

PAD LOCATION

Pad No.	Pad name	X	Y	Pad No.	Pad name	X	Y
1	SC51	-1313.50	1746.00	51	SC1	1542.00	-1754.00
2	SC52	-1544.00	1746.00	52	SC2		-1629.00
3	SC53		1621.00	53	SC3		-1504.00
4	SC54		1496.00	54	SC4		-1379.00
5	SC55		1371.00	55	SC5		-1254.00
6	SC56		1246.00	56	SC6		-1129.00
7	SC57		1121.00	57	SC7		-1004.00
8	SC58		996.00	58	SC8		-879.00
9	SC59		871.00	59	SC9		-754.00
10	SC60		746.00	60	SC10		-629.00
11	SC61		621.00	61	SC11		-504.00
12	SC62		496.00	62	SC12		-379.00
13	SC63		371.00	63	SC13		-254.00
14	SC64		246.00	64	SC14		-129.00
15	SC65		121.00	65	SC15		-4.00
16	SC66		-4.00	66	SC16		121.00
17	SC67		-129.00	67	SC17		246.00
18	SC68		-254.00	68	SC18		371.00
19	SC69		-379.00	69	SC19		496.00
20	SC70		-504.00	70	SC20		621.00
21	SC71		-629.00	71	SC21		746.00
22	SC72		-754.00	72	SC22		871.00
23	SC73		-879.00	73	SC23		996.00
24	SC74		-1004.00	74	SC24		1121.00
25	SC75		-1129.00	75	SC25		1246.00
26	SC76		-1254.00	76	SC26		1371.00
27	SC77		-1379.00	77	SC27		1496.00
28	SC78		-1504.00	78	SC28		1621.00
29	SC79		-1629.00	79	SC29		1746.00
30	SC80		-1754.00	80	SC30	1311.50	
31	ERB	-1218.40		81	SC31	1186.50	
32	VEE	-1048.70		82	SC32	1061.50	
33	V5	-923.70		83	SC33	936.50	
34	V43	-798.70		84	SC34	811.50	
35	V12	-673.70		85	SC35	686.50	
36	V0	-548.70		86	SC36	561.50	
37	CS	-380.00		87	SC37	436.50	
38	M	-255.00		88	SC38	311.50	
39	DISPOFFB	-130.00		89	SC39	186.50	
40	VDD	-5.00		90	SC40	61.50	
41	SHL	120.10		91	SC41	-63.50	
42	VSS	245.10		92	SC42	-188.50	
43	D4_DR	370.10		93	SC43	-313.50	
44	D3_DM	495.10		94	SC44	-438.50	
45	D2_DL	620.10		95	SC45	-563.50	
46	D1_SID	745.10		96	SC46	-688.50	
47	CL2	870.10		97	SC47	-813.50	
48	AMS	995.10		98	SC48	-938.50	
49	CL1	1120.10		99	SC49	-1063.50	
50	ELB	1245.10		100	SC50	-1188.50	

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BLOCK DIAGRAM



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BLOCK DESCRIPTION

NAME	FUNCTION	COM / SEG
Clock control	Generates latch clock (LCK), shift clock (SCK) and control clock timing according to the input of CL1, CL2 and control inputs (CS, AMS). In common driver application mode, this block generates the shift clock (LCK) for the common data Bi-directional shift register.	COM / SEG
Data latch control	Determines the direction of segment data shift, and input data of each Data latch Bi-directional shift register. In 4-bit segment data parallel transfer mode, data is shifted by a 4-bit unit. In common driver application mode, data is transferred to the common data shift register directly, which disables this block.	SEG
Power down function	Controls the clock enable state of the current driver according to the input value of enable pin (ELB or ERB). If enable input value is “Low”, every clock of the current driver is enabled and the clock control block works. But if enable input is “High”, current driver is disabled and the input data value has no effect on the output level. So power consumption can be lowered.	SEG
Output level selector	Controls the output voltage level according to the input control pin (M and DISPOFFB) (refer to PIN DESCRIPTION).	COM / SEG
20x4-bitsegment data bi-directional shift register	Stores output data value by shifting the input values. In 1-bit serial interface mode application, all 80 shift clocks (SCK) are needed to store all the display data. But in 4-bit parallel transfer mode application, only 20 clocks are needed. In common driver application mode, this block does not work.	SEG
80-bit data latch / common data bi-directional shift register	In segment driver application mode, the data from the 20x4-bit segment data shift register are latched for segment driver output. In single-type common driver application, 1-bit input data (from DL or DR pin) is shifted and latched by the direction according to the SHL signal input. In dual-type common application mode, 80-bit registers are divided by two blocks and controlled Independently (refer to NOTE 3).	COM / SEG
80-bit level shifter	Voltage level shifter block for high voltage part. The inputs of this block are of logical voltage level and the outputs of this block are at high voltage level value. These values are input in to the driver.	SEG
80-bit 4-level driver	Selects the output voltage level according to M and latched data value. If the data value is "High" the driver output is at selected voltage level (V0 or V5), and in the reverse case the driver output value is at the non-selected level (V12 or V43). In segment driver application mode, non-selected output value is V2 or V3 and when in common driver application, this value becomes V1 or V4.	SEG

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PIN DESCRIPTION

PIN	I/O	NAME	FUNCTION	INTERFACE
VDD		Power supply	Logical "High" input port (+5V±10%, +3V± 10%)	Power
VSS			0V (GND)	
VEE			Logical "Low" for high voltage part	
V0,V12, V43,V5	I	LCD driver output voltage level	Bias supply voltage input to drive the LCD. Bias voltage divided by the resistance is usually used as a supply voltage source (refer to NOTE 2).	Power
SC1~SC80	O	LCD driver output	Display data output pin which corresponds to the respective latch contents. One of V0, V12, V34 and V5 is selected as a display driving voltage source according to the combination of the latched data level and M signal (refer to NOTE 1).	LCD
CL2	I	Data shift clock	Clock pulse input for the bi-directional shift register. – **In segment driver application mode**, the data is shifted to 20 x4-bit segment data shift. The clock pulse, which was input when the enable bit (ELB/ERB) is in not active condition, is invalid. – **In common driver application mode**, the data is shifted to 80-bit common data bi-directional shift register by the CL1 clock. Hence, this clock pin is not used (Open or connect this pin to VDD).	Controller
M	I	AC signal for LCD driver output	Alternate signal input pin for LCD driving. Normal frame inversion signal is input in to this pin.	Controller
CL1	I	Data latch clock	– **In segment driver application mode**, this signal is used for latching the shift register contents at the falling edge of this clock pulse. CL1 pulse "High" level initializes power-down function block. – **In common driver application mode**, CL1 is used as a shifting clock of common output data.	Controller

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PIN DESCRIPTION (CONTINUED)

<table>
<tr><th>PIN</th><th>I/O</th><th>NAME</th><th>FUNCTION</th><th>INTERFACE</th></tr>
<tr><td>DISPOFFB</td><td>I</td><td>Display OFF control</td><td>Control input pin to fix the driver output (SC1~SC80) to V0 level, during "Low" value input. LCD becomes non-selected by V0 level output from every output of segment drivers and every output of common drivers.</td><td>Controller</td></tr>
<tr><td>CS</td><td>I</td><td>COM / SEG mode control</td><td>When CS = "Low", NT7086PQ is used as an 80-bit segment driver.
When CS = "High", NT7086PQ is set to an 80-bit common driver</td><td>VDD/VSS</td></tr>
<tr><td>AMS</td><td>I</td><td>Application mode select</td><td>According to the input value of the AMS and the CS pin, application mode of NT7086PQ is differs as shown below.
<table>
<tr><th>CS</th><th>AMS</th><th>Application mode</th><th>COM/SEG</th></tr>
<tr><td>0</td><td>0</td><td>4-bit parallel interface mode</td><td rowspan="2">SEG</td></tr>
<tr><td>0</td><td>1</td><td>1-bit serial interface mode</td></tr>
<tr><td>1</td><td>0</td><td>Single type application Mode</td><td rowspan="2">COM</td></tr>
<tr><td>1</td><td>1</td><td>Dual type application mode</td></tr>
</table></td><td>VDD/VSS</td></tr>
<tr><td>D1_SID, D2_DL, D3_DM, D4_DR</td><td>I/O</td><td>Display data input/ serial input data/ left , right data input output</td><td>-In segment driver mode, these pins are used as 4-bit data input pin (when 4-bit parallel interface mode AMS= ”low”), or D1_SID is used as serial data input pin and other pins are not used (connect these to VDD) (when 1-bit serial interface mode AMS= ”high”).
-In common driver mode, the data is shifted from D2_DL (D4_DR) to D4_DR (D2_DL), when in single interface mode (AMS= ”Low”). In dual-type application case, the data are shifted from D2_DL and D3_DM (D4_DR and D3_DM) to D4_DR(D2_DL). In each case the direction of the data shift and the connection of data pins are determined by SHL input (refer to NOTE 3, NOT 4).</td><td>Controller</td></tr>
</table>

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PIN DESCRIPTION (CONTINUED)

PIN	I/O	NAME	FUNCTION	INTERFACE
SHL	I	Shift direction control	When SHL = "Low", data is shifted from left to right. When SHL = "High", the direction is reversed. (refer to NOTE3)	VDD/VSS
ELB, ERB	I/O	Enable data input/ output	-**In segment driver mode**, the internal operation is enabled only when enable input (ELB or ERB) is "Low" (power down function). When several drivers a serially connected, the enable state of each driver is shifted according to the SHL input. Connect these pins as below. **SHL** / **Segment driver** / **ELB** / **ERB** L / Output (open) / Input (VSS) H / Input (VSS) / Output(open) -**In common driver mode**, the power down function is not used. Open these pins.	

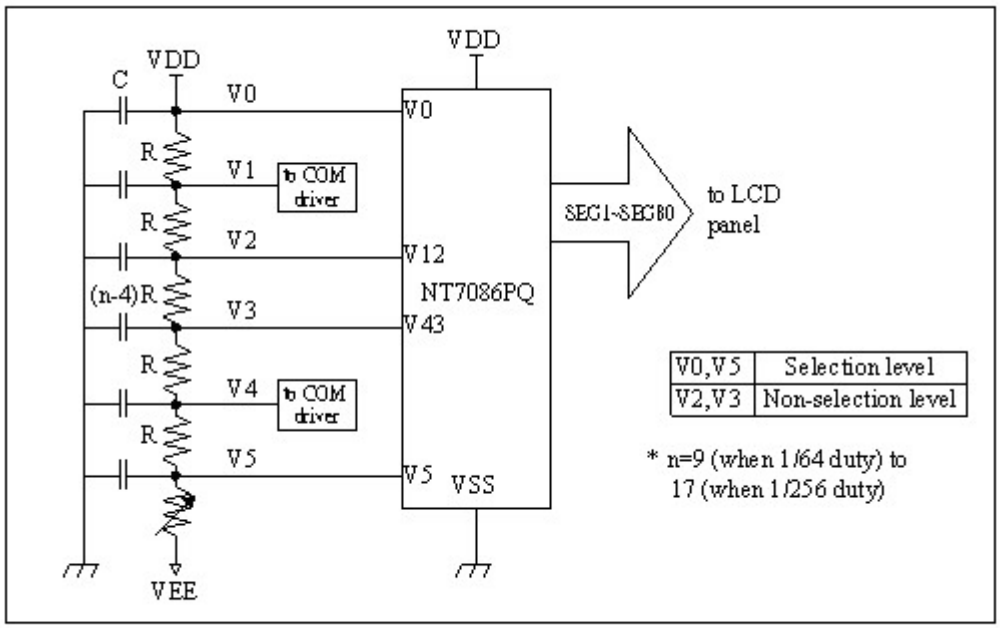
NOTE 1. Output level control **"X": don't care**

M	Latched data	DISPOFFB	Output level (CS1~CS80)	
			SEG Mode	COM Mode
L	L	H	V12(V2)	V12(V1)
L	H	H	V0	V5
H	L	H	V43(V3)	V43(V4)
H	H	H	V5	V0
X	X	L	V0	V0

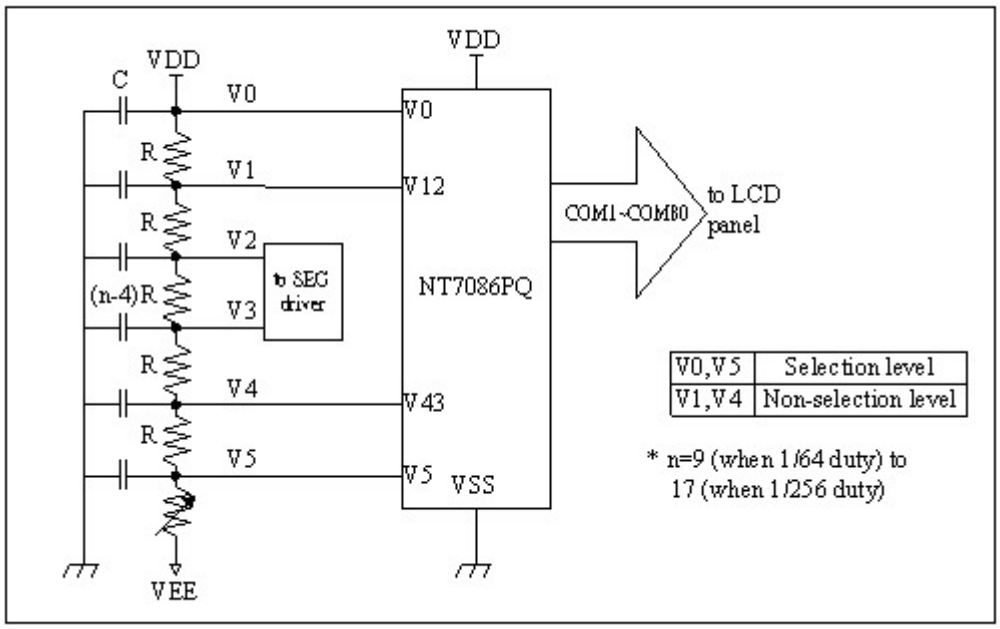
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NOTE 2. LCD Driving Voltage Application Circuit

(1) Segment driver application (CS = "Low")



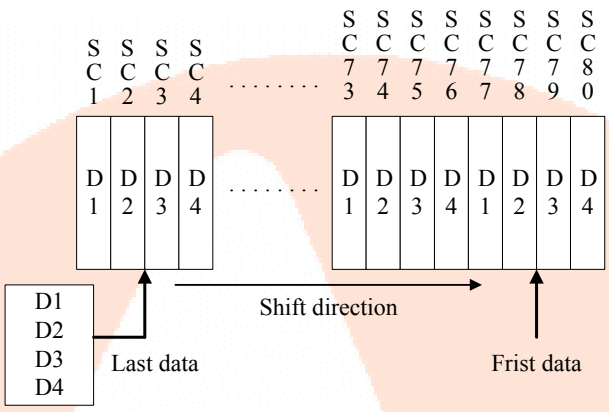
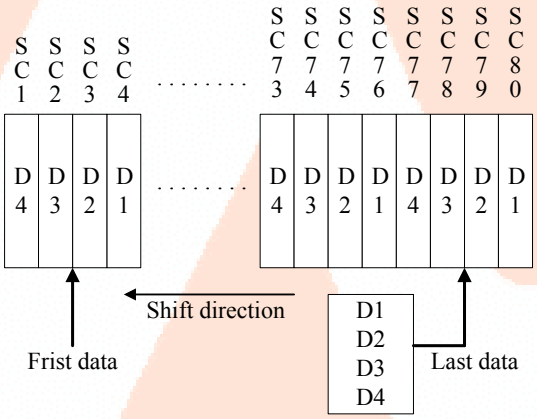
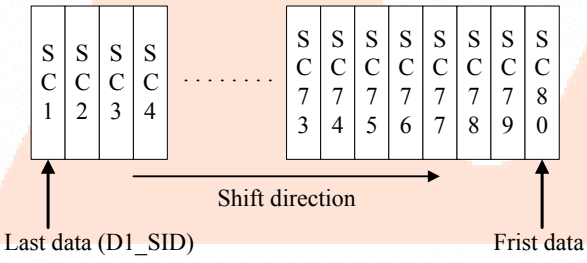
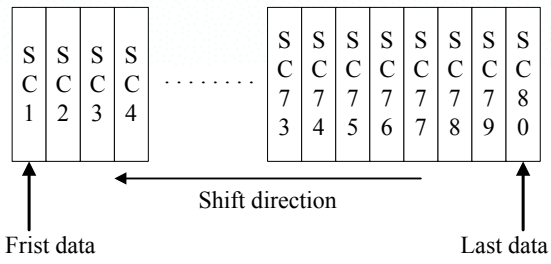
(2) Common driver application (CS = "High")



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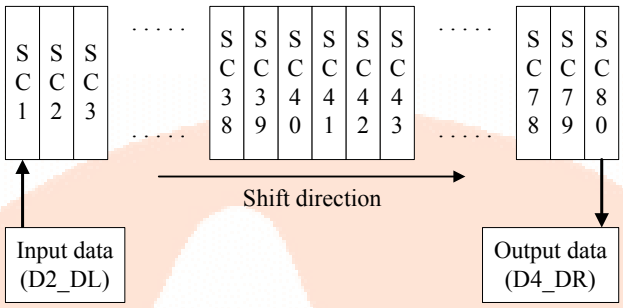
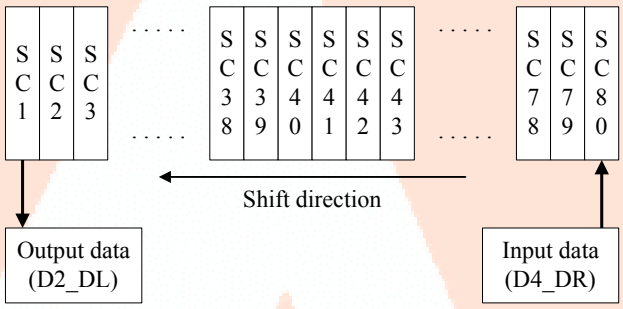
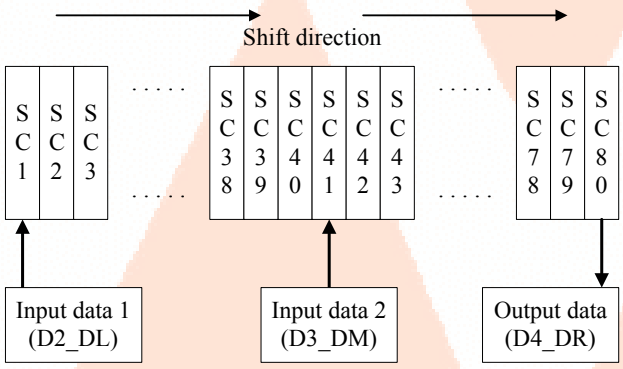
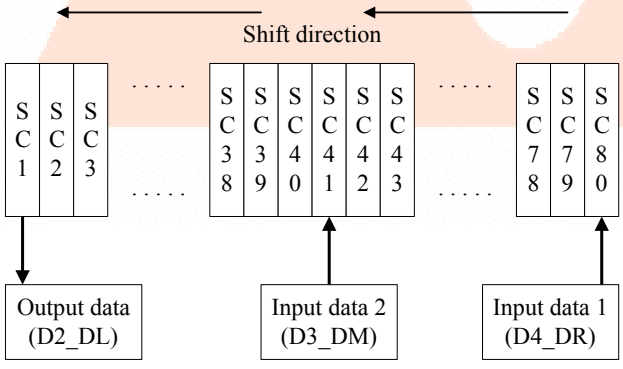
NOTE 3. Data Shift Direction according to Control Signals

(1) When CS = "Low" (segment driver application)

<table>
<tr><th>AMS</th><th>SHL</th><th>Application mode</th><th>Data Direction</th><th>Input pin</th></tr>
<tr><td rowspan="2">L</td><td>L</td><td rowspan="2">4-bit parallel data transfer mode (SEG)</td><td> 
</td><td rowspan="2">D1_SID,
D2_DL,
D3_DM,
D4_DR</td></tr>
<tr><td>H</td><td> 
</td></tr>
<tr><td rowspan="2">H</td><td>L</td><td rowspan="2">1-bit serial data transfer mode (SEG)</td><td> 
</td><td rowspan="2">D1_SID</td></tr>
<tr><td>H</td><td> 
</td></tr>
</table>

Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

(2) When CS = "High" (common driver application)

AMS	SHL	Application mode	Data Direction	Input pin
L	L	Single-type Application mode (COM)		D2_DL
	H			D4_DR
H	L	Dual-type Application mode (COM)		D2_DL, D3_DM
	H			D4_DR, D3_DM

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NOTE 4. Usage of Data Pins

COM / SEG (CS pin)	Application mode (AMS pin)	SHL	Data interface pin			
			D1_SID	D2_DL	D3_DM	D4_DR
SEG (CS ="Low")	4-bit parallel interface mode (AMS = "Low")	X	D1 (input)	D2 (input)	D3 (input)	D4(input)
	1-bit serial interface mode (AMS = "High")	X	SID (input)		Connect to V_{DD}	
COM (CS ="High")	single-type application mode (AMS = "Low")	L	open	DL (input)	Open	DR (output)
		H		DL (output)		DR (input)
	dual-type application mode (AMS = "High")	L	open	DL (input1)	DM (input2)	DR (output2)
		H		DL (output2)	DM (input2)	DR (input1)

MAXIMUM ABSOLUTE LIMIT

Characteristic	Symbol	Value	Unit
Power supply voltage	V_{DD}	-0.3~+7.0	V
Driver supply voltage	V_{LCD}	0~+30	
Input voltage	V_{IN}	-0.3~V_{DD}+0.3	
Operating temperature	Topr	-30~+85	℃
Storage temperature	Tstg	-55~+150	

NOTE: Voltage greater than above may do damage to the circuit.

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ELECTRICAL CHARACTERISTICS

DC CHARACTERISTICS

(1) Segment Driver Application

(V_{SS} = 0V, Ta = - 30 ~ +85°C)

Characteristic	Symbol	Test Condition		Min.	Typ.	Max.	Unit
Operating Voltage 1	V_{DD}	-		2.7	-	5.5	V
	V_{LCD}	V_{IN}=V_{DD}-V_{EE}		6	-	28	
Input voltage (1)	V_{IH}	-		0.8V_{DD}	-	V_{DD}	
	V_{IL}	-		0	-	0.2V_{DD}	
Input voltage (2)	V_{OH}	I_{CH}=-0.4mA		V_{DD}-0.4	-	-	V
	V_{OL}	I_{OH}=-0.4mA		-	-	0.4	
Input leakage current 1 (1)	I_{IL1}	V_{IN}=V_{DD} to V_{SS}		-10	-	10	μA
Input leakage current 2 (3)	I_{IL2}	V_{IN}=V_{DD} to V_{EE}		-25	-	25	
On resistance(4)	R_{ON}	I_{ON}=100 μA		-	2	4	kΩ
Supply current(5)	I_{STBY}	f_{CL1}=32kHZ, M=VSS	V_{SS} PIN	-	-	100	μA
	I_{DD}	f_{CL1}=32kHZ F_M=80HZ	V_{DD}=5V	-	-	5	mA
			V_{DD}=3V	-	-	2	
	I_{EE}		V_{DD}=5V	-	-	500	μA

NOTES:

1. Applied to CL1, CL2, ELB, ERB, D1_SID - D4_DR, SHL, DISPOFFB, M, CS, AMS pin
2. ELB, ERB pin
3. V0, V12, V43, V5 pin
4. V_{LCD} = V_{DD} - V_{EE}, V0 = V_{DD} = 5V, V5= V_{EE} = -23 V
V12 = V_{DD}-2/n(V_{LCD}), V43 = V_{EE}+2/n(V_{LCD}), n = 17 (1/256 duty, 1/17 bias)
5. V0 = V_{DD}, V12 = 1.71V(V_{DD} = 5V) or -0.06V (V_{DD} = 3V),
V43 = -19.71 V(VDD = 5V) or -19.94V (V_{DD} = 3V), V5 = V_{EE} = -23V, no-load condition (1/256 duty, 1/17 bias)
4-bit parallel interface mode
I_{STBY} : V_{DD} = 5V, f_{CL2} = 5.12MHz, SHL = V_{SS}, DISPOFFB = V_{DD}, M = V_{SS}, display data pattern = 0000
I_{DD} : V_{DD} = 3V, f_{CL2} = 4MHz, display data pattern = 0101
V_{DD} = 5 V, f_{CL2} = 5.12MHz, display data pattern = 0101
I_{EE} : V_{DD} = 5V, f_{CL2} = 5.12MHz, display data pattern = 0101, V_{EE} pin

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DC CHARACTERISTICS (CONTINUED)
(2) Common Driver Application

(V_{SS} = 0V, Ta = - 30 ~ +85°C)

Characteristic	Symbol	Test Condition		Min.	Typ.	Max.	Unit
Operating Voltage 1	V_{DD}	-		2.7	-	5.5	V
	V_{LCD}	$V_{IN}=V_{DD}-V_{EE}$		6	-	28	
Input voltage (1)	V_{IH}	-		$0.8V_{DD}$	-	V_{DD}	
	V_{IL}	-		0	-	$0.2V_{DD}$	
Input voltage (3)	V_{OH}	I_{CH}=-0.4mA		V_{DD}-0.4	-	-	V
	V_{OL}	I_{OH}=-0.4mA		-	-	0.4	
Input leakage current 1 (1)	I_{IL1}	$V_{IN}=V_{DD}$ to V_{SS}		-10	-	10	μA
Input leakage current 2 (2)	I_{IL2}	V_{IN}=0V, V_{DD}=5V(Pull up)		-50	-125	-250	
Input leakage current 3 (4)	I_{IL3}	$V_{IN}=V_{DD}$ to V_{EE}		-25	-	25	
On resistance(5)	R_{ON}	I_{ON}=100 μA		-	2	4	kΩ
Supply current(6)	I_{STBY}	f_{CL1}=32kHZ, M=VSS	V_{SS} PIN	-	-	100	μA
	I_{DD}	f_{CL1}=32kHZ F_M=80HZ	V_{DD}=5V	-	-	200	
			V_{DD}=3V	-	-	120	
	I_{EE}		V_{DD}=5V	-	-	150	

NOTES:

1. Applied to CL1, D2_DL (SHL = LOW), D4_DR (SHL = HIGH), SHL, DISPOFFB, M, CS, AMS pin
2. Pull-up input pins : CL2, D1_SID, D3_DM (AMS = HIGH), ELB (SHL = LOW), ERB (SHL = HIGH)
3. D2_DL (SHL = HIGH) , D4_DR (SHL = LOW) pin
4. V0, V12, V43, V5 pin
5. $V_{LCD} = V_{DD}-V_{EE}$, V0 = V_{DD} = 5V, V5 = V_{EE} = -23V
V12 = V_{DD}-1/n(V_{LCD}), V43 = V_{EE}+1/n(V_{LCD}), n = 17(1/256 duty, 1/17 bias)
6. V0 = V_{DD}, V12 = 3.35V (V_{DD} = 5V) or 1.47V (V_{DD} = 3V),
V43 = -21.35V (V_{DD} = 5 V) or -21.47V (V_{DD} = 3 V), V5 = V_{EE} = -23 V, no-load condition (1/256 duty, 1/17 bias)
single-type mode operation : AMS = V_{SS}, SHL = V_{SS}, DISPOFFB = V_{DD}
D1_SID = D3_DM = VDD, D4_DR = OPEN, ELB = ERB = OPEN,
I_{STBY} : V_{DD} = 5V, M = V_{SS}, D2_DL = V_{SS}
I_{DD} : f_M = 80Hz, D2_DL = V_{DD}
V_{DD} = 3 V, display data pattern = 10000000..., 01000000..., 00100000..., 00010000..., ..
V_{DD} = 5 V, display data pattern = 10000000..., 01000000..., 00100000..., 00010000..., ..
I_{EE} : f_M = 80Hz, D2_DL = V_{DD}
V_{DD} = 5V, current through V_{EE} Pin, display data pattern = 10000000..., 01000000...,
00100000..., 00010000...

Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

AC CHARACTERISTICS

(1) Segment Driver Application

(V_{SS} = 0V, Ta = - 30 ~ +85°C)

Characteristic	Symbol	Test condition	(1) VDD=5V±10%			(2) VDD=3V±10%			Unit
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Clock cycle time	t_{CY}	Duty=50%	125	-	-	250	-	-	ns
Clock pulse width	t_{WCK}	-	45	-	-	95	-	-	
Clock rise/ fall time	t_R / t_F	-	-	-	-	-	-	30	
Data set-up time	t_{DS}	-	30	-		65	-	-	
Data hold time	t_{DH}	-	30	-		65	-	-	
Clock set-up time	t_{CS}	-	80	-		120	-	-	
Clock hold time	t_{CH}	-	80	-		120	-	-	
Propagation delay time	t_{PHL}	ELB output	-	-	60	-	-	125	
		ERB output			60			125	
ELB,ERB set-up time	t_{PSU}	ELB input	30	-	-	65	-	-	
		ERB input	30			65			
DISPOFFB low pulse width	t_{WDL}	-	1.2	-	-	1.2	-	-	μs
DISPOFFB clear time	t_{CD}	-	100	-	-	100	-	-	ns
M – OUT propagation delay time	t_{PD1}	C_L=15pF	-	-	1.0	-	-	1.2	μs
CL1 – OUT propagation delay time	t_{PD2}		-	-	1.0	-	-	1.2	
DISPOFFB – OUT propagation delay time	t_{PD3}		-	-	1.0	-	-	-	

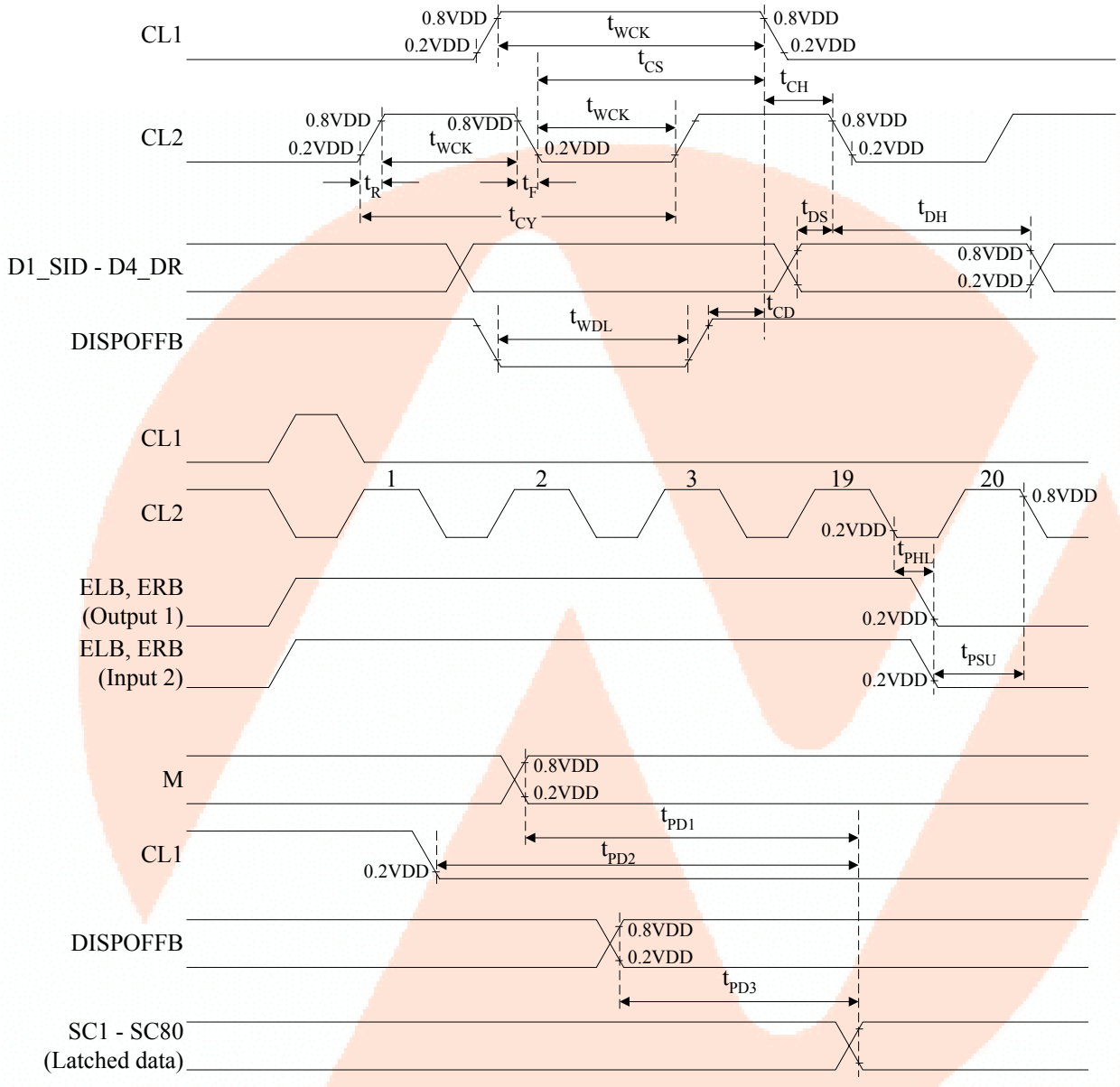
(2) Common Driver Application

(V_{SS} = 0V, Ta = - 30 ~ +85°C)

Characteristic	Symbol	Test condition	(1) VDD=5V±10%			(2) VDD=3V±10%			Unit
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Clock cycle time	t_{CY}	Duty=50%	250	-	-	500	-	-	ns
Clock pulse width	t_{WCK}	-	45	-	-	95	-	-	
Clock rise/ fall time	t_R / t_F	-	-	-	50	-	-	50	
Data set-up time	t_{DS}	-	30	-		65	-	-	
Data hold time	T_{DH}	-	30	-		65	-	-	
DISPOFFB low pulse width	t_{WDL}	-	1.2	-	-	1.2	-	-	μs
DISPOFFB clear time	t_{CD}	-	100	-	-	100	-	-	ns
Output delay time	t_{DL}	C_L=15pF	-	-	200	-	-	250	
M – OUT propagation delay time	t_{PD1}		-	-	1.0	-	-	1.2	μs
CL1 – OUT propagation delay time	t_{PD2}		-	-	1.0	-	-	1.2	
DISPOFFB – OUT propagation delay time	t_{PD3}		-	-	1.0	-	-	1.2	

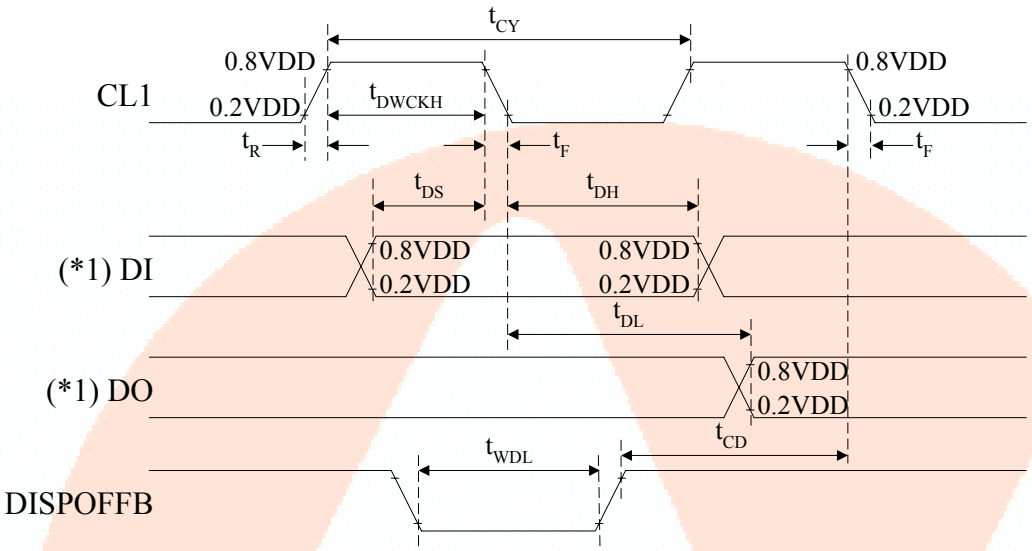
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(3) Segment Driver Application Timing

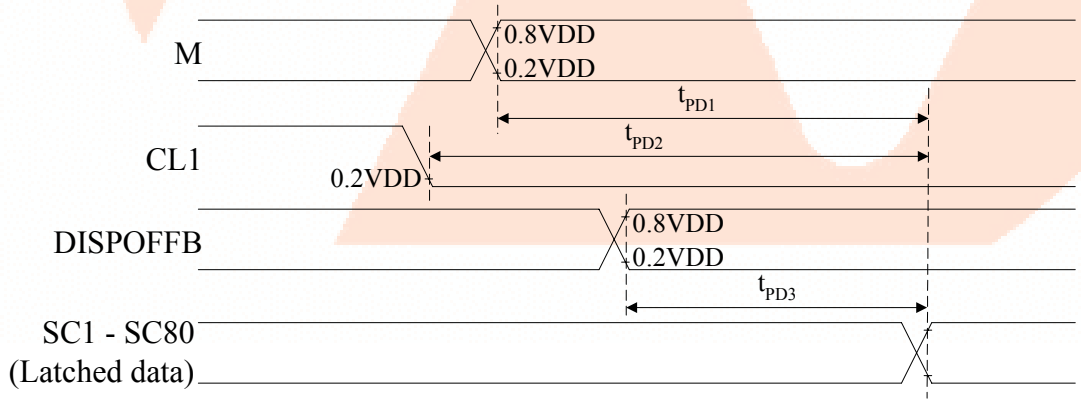


Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

(4) Common Driver Application Timing



(*1) When in single-type interface mode
DI=>DDL(SHL=L), D4_DR(SHL=H)
DO=>D4_DR(SHL=L), D2_DL(SHL=H)
When in dual-type interface mode
DI=>D2_DL and D3_DM(SHL=L),D4_DR and D3_DM(SHL=H)
DO=>D4_DR(SHL=L), D2_DL(SHL=H)



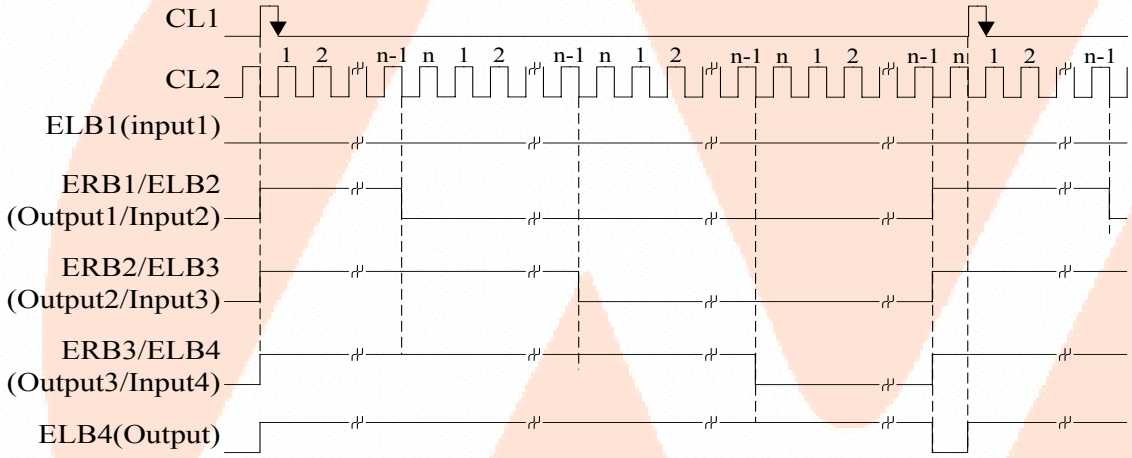
Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

POWER DOWN FUNCTION

In the case of cascade connection of segment mode drivers, NT7086PQ has a "power down function" In order to reduce the power consumption.

SHL	Enable input	Enable output	Current driver status	The other drivers status
L	ERB	ELB	While ERB ="Low", current driver is enabled.	Disabled
H	ELB	ERB	While ELB ="Low", current driver is enabled.	Disabled

* In the case of common driver application, power down function does not work.



NOTES:

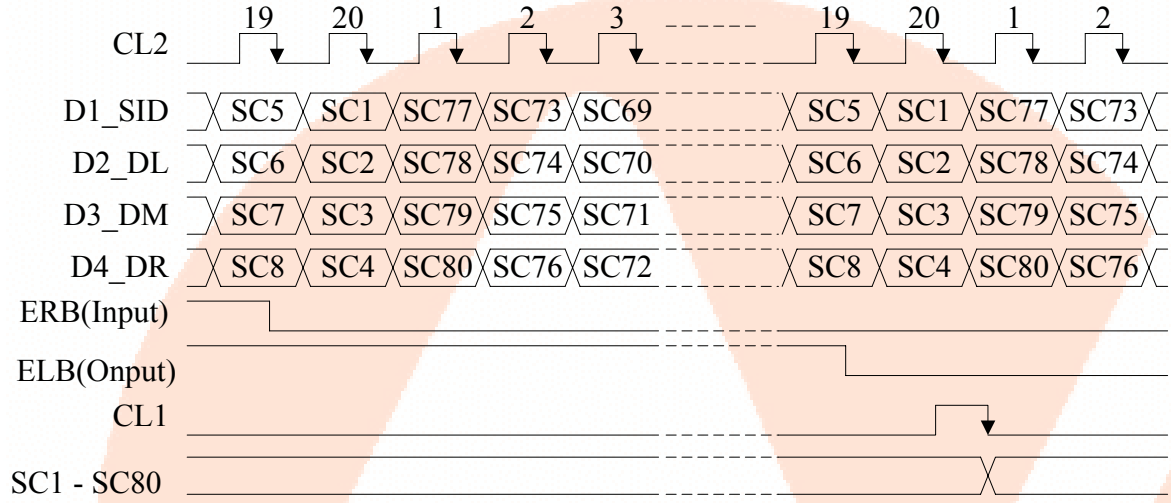
1. SHL = High (ELB = Input, ERB = Output)
2. When in 4-bit parallel interface mode: n = 20
   When in 1-bit serial interface mode: n = 80

Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

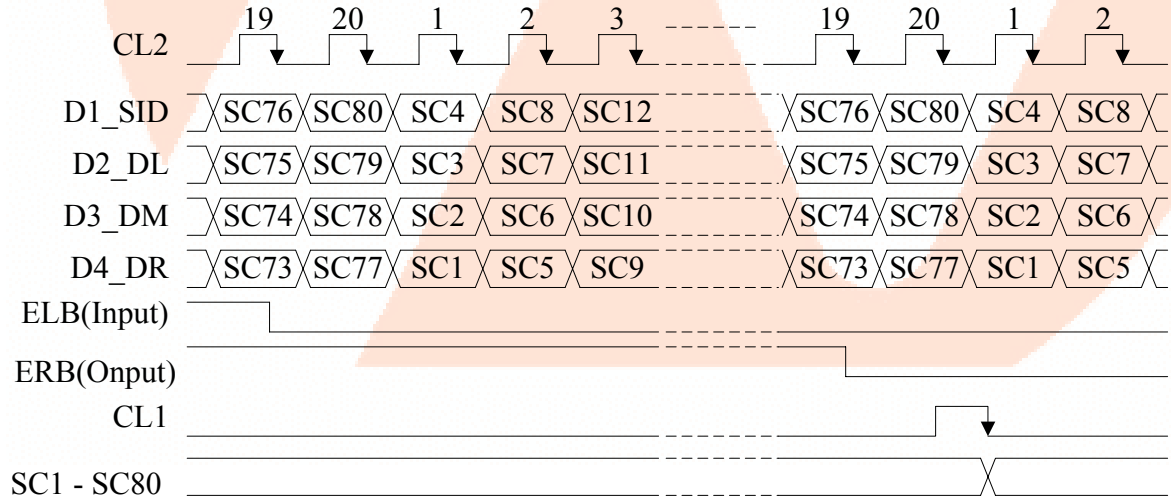
OPERATION TIMING DIAGRAM

(1) 4-bit parallel mode interface segment driver

When SHL= ”Low”



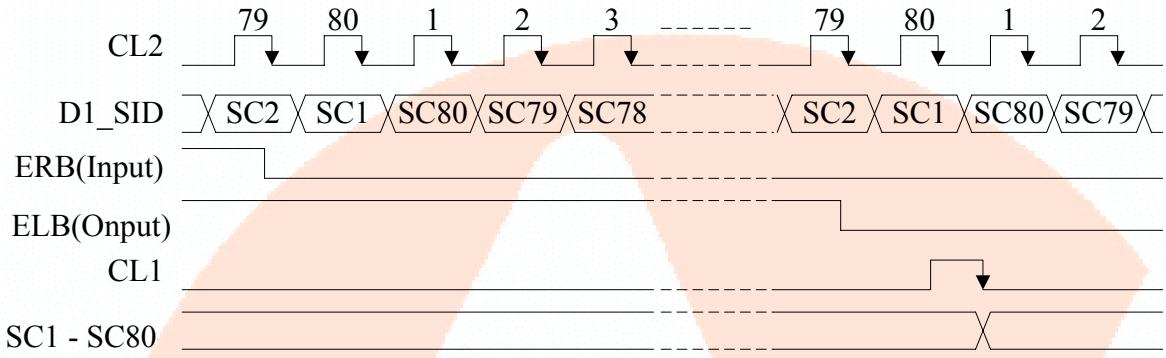
When SHL= ”High”



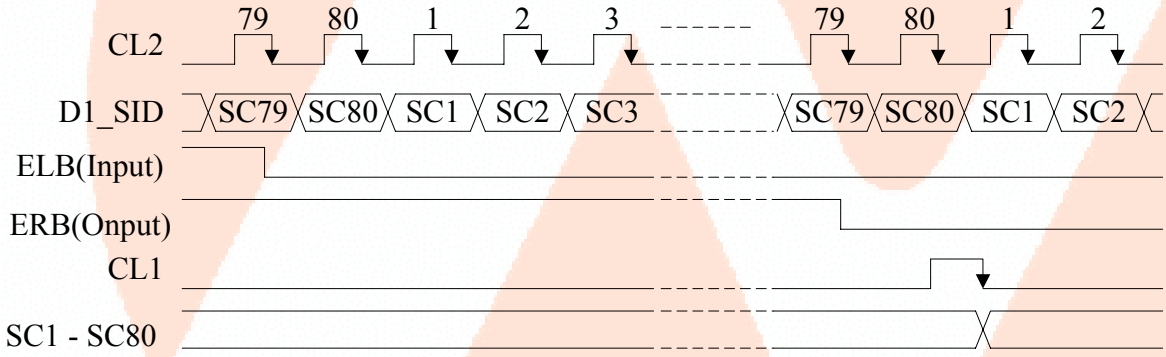
Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

(2) 1-bit serial mode interface segment driver

When SHL= ”Low”



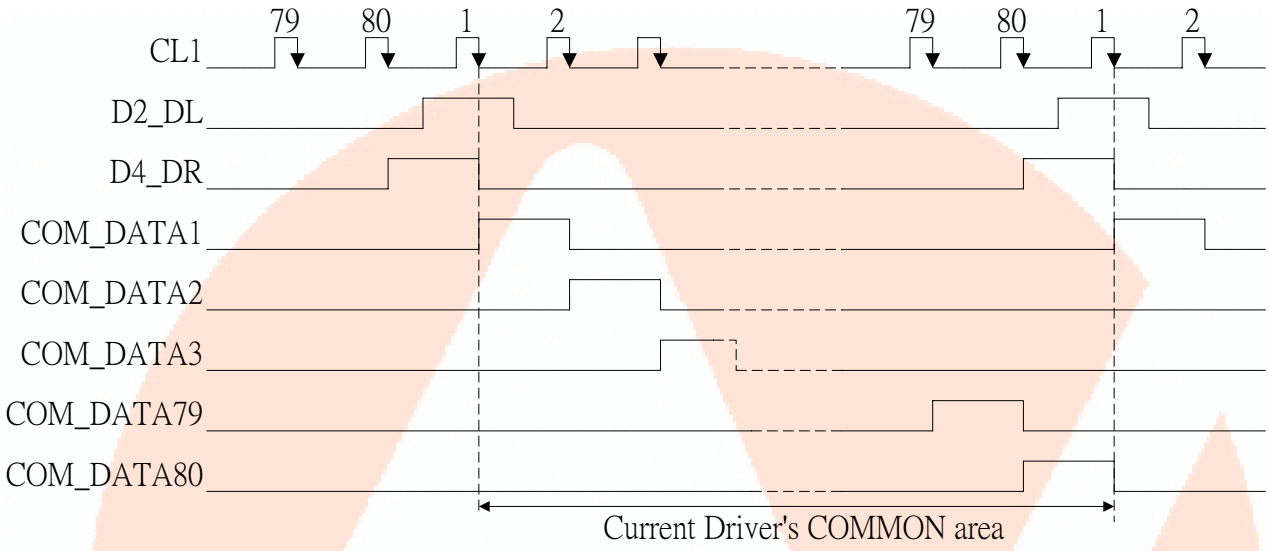
When SHL= “High”



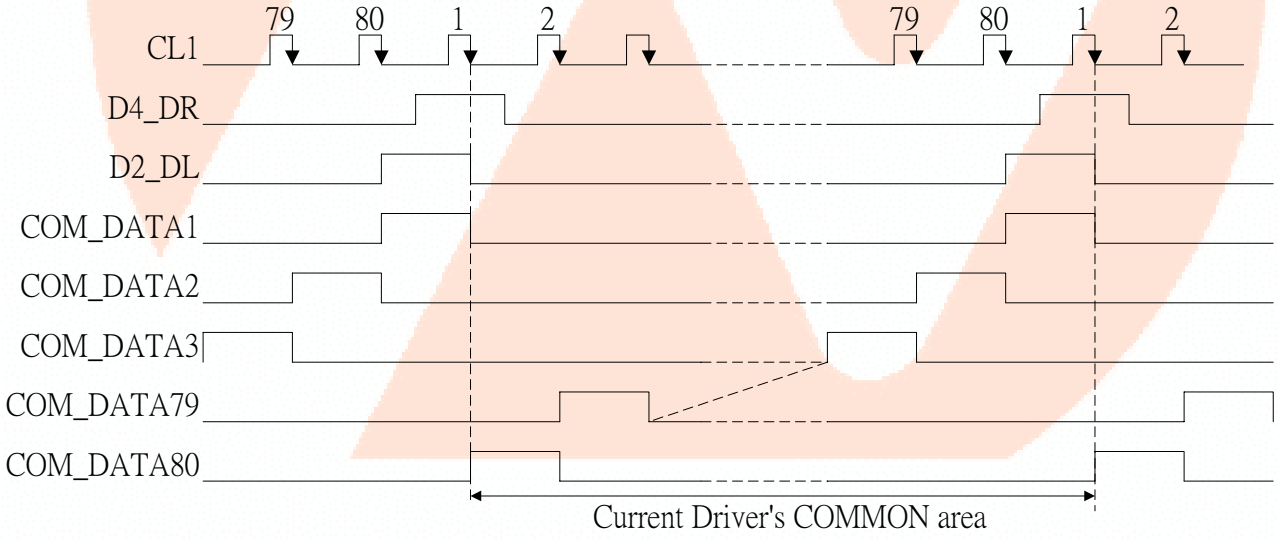
Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

(3) Single type interface mode common driver

When SHL= “Low”



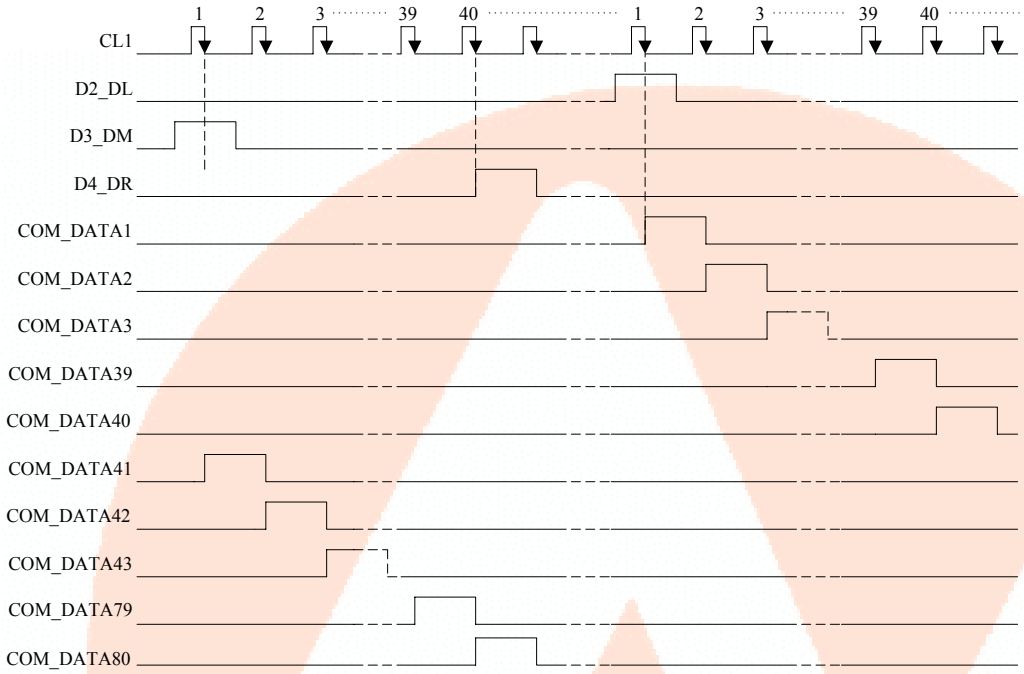
When SHL= ”High”



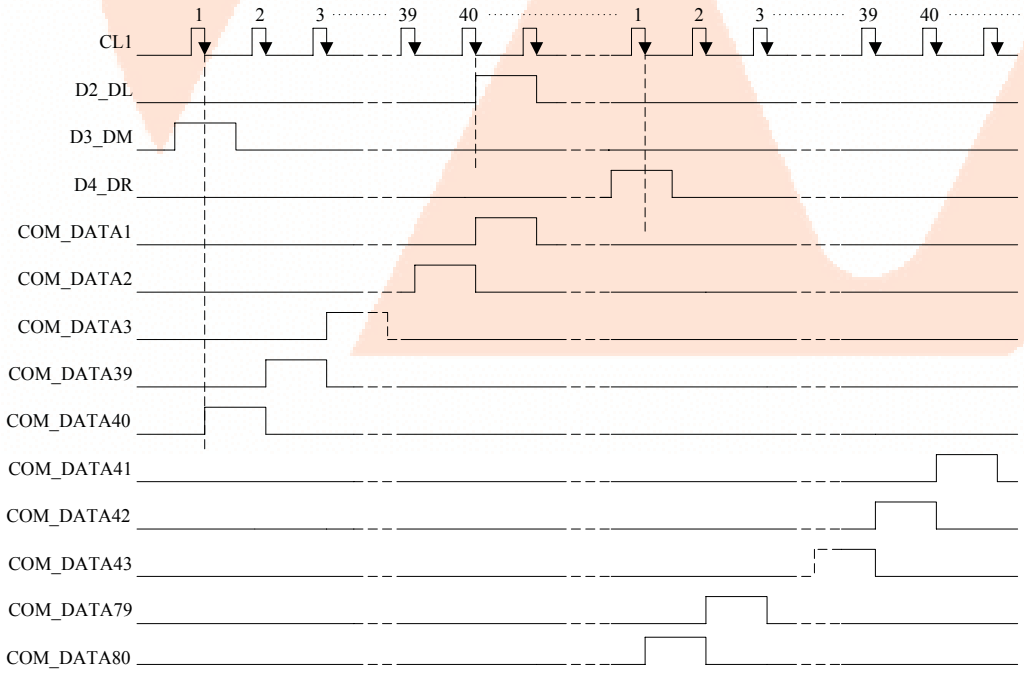
Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

(4) Dual-type interface mode common driver

When SHL= “Low”

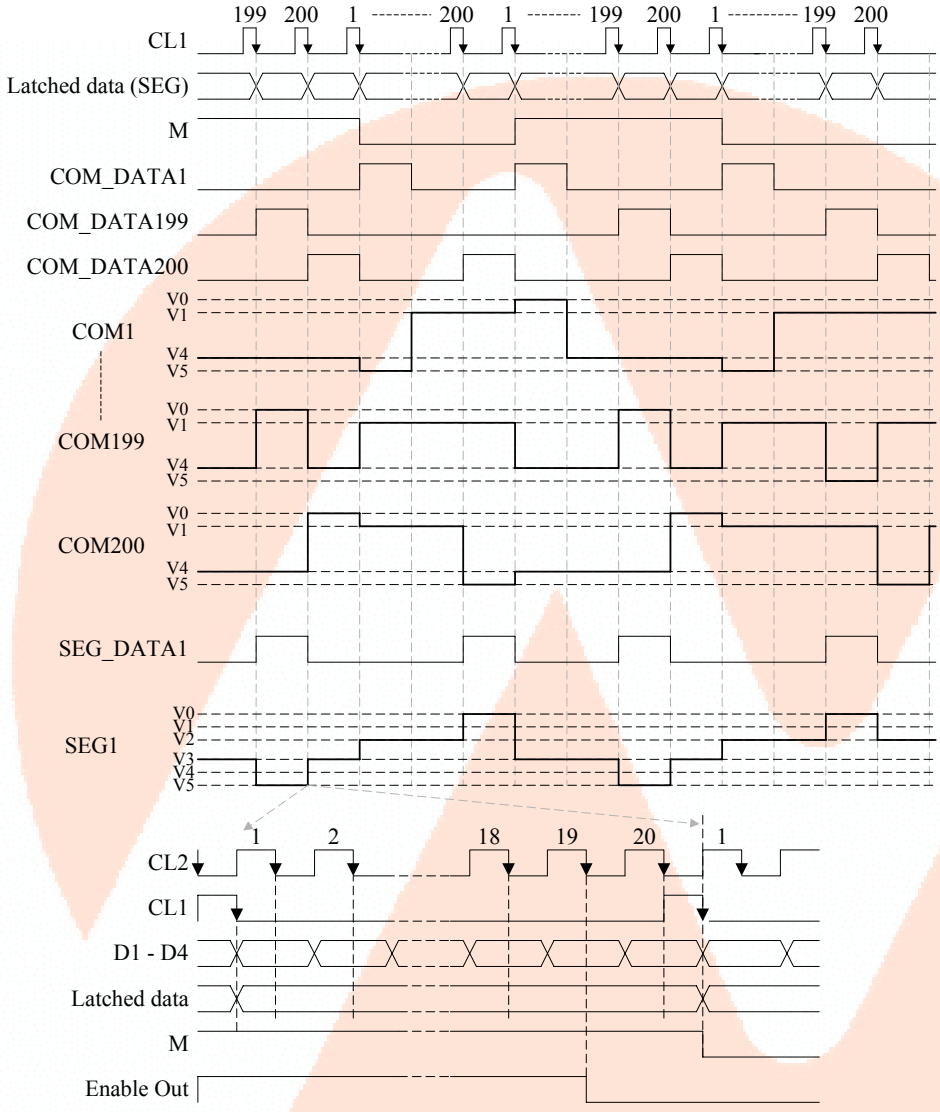


When SHL= “High”



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(5) Common / Segment driver timing (1/200 duty)

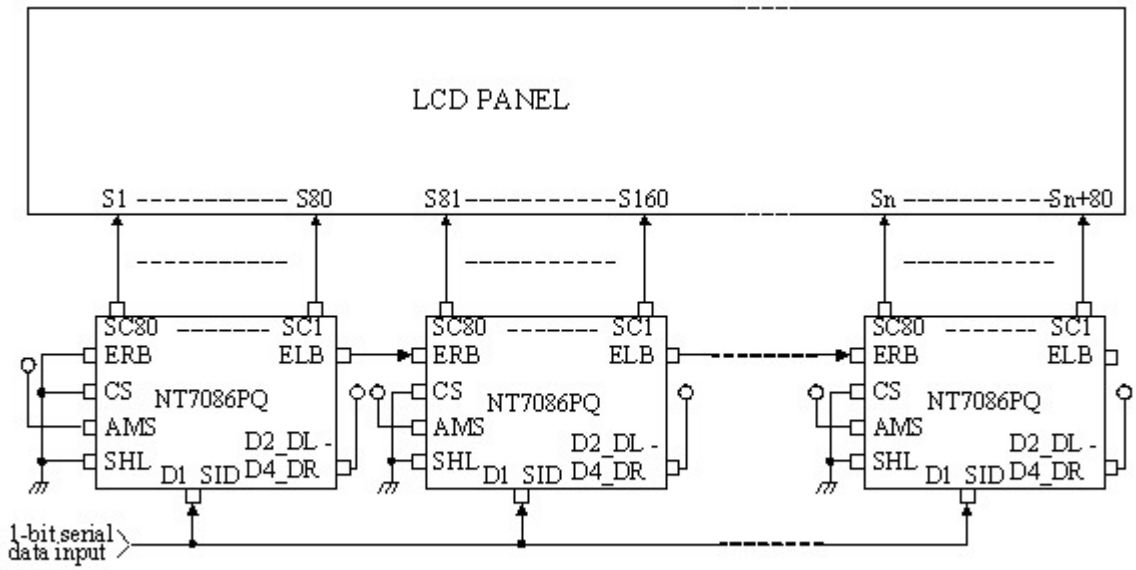


Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

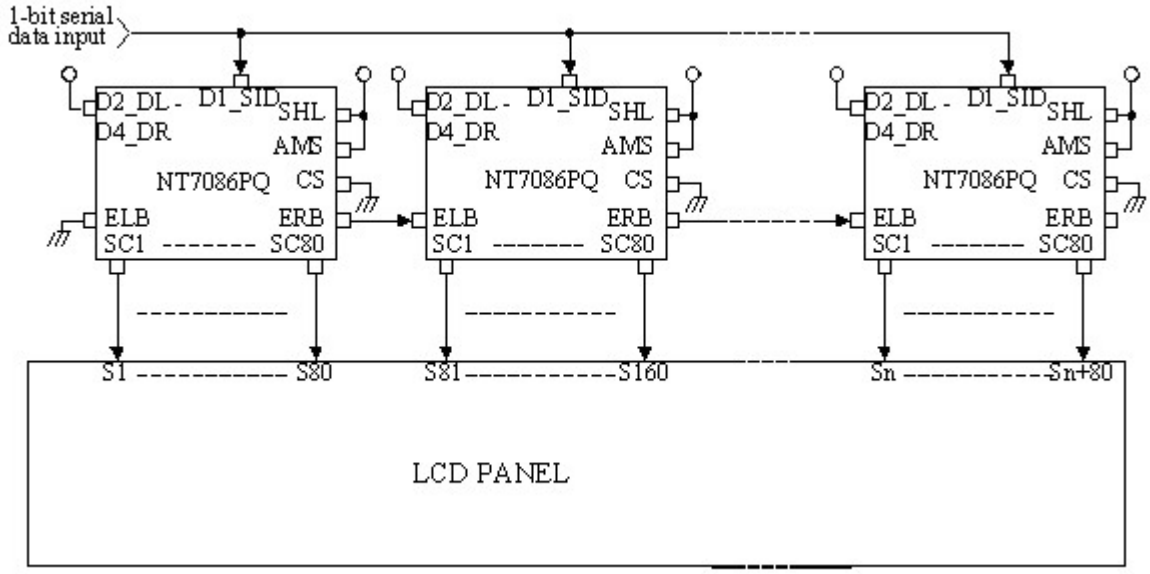
APPLICATION INFORMATION

1-bit serial interface mode (80 Ch. Segment mode)

a) Lower view (SHL= L, AMS= H)



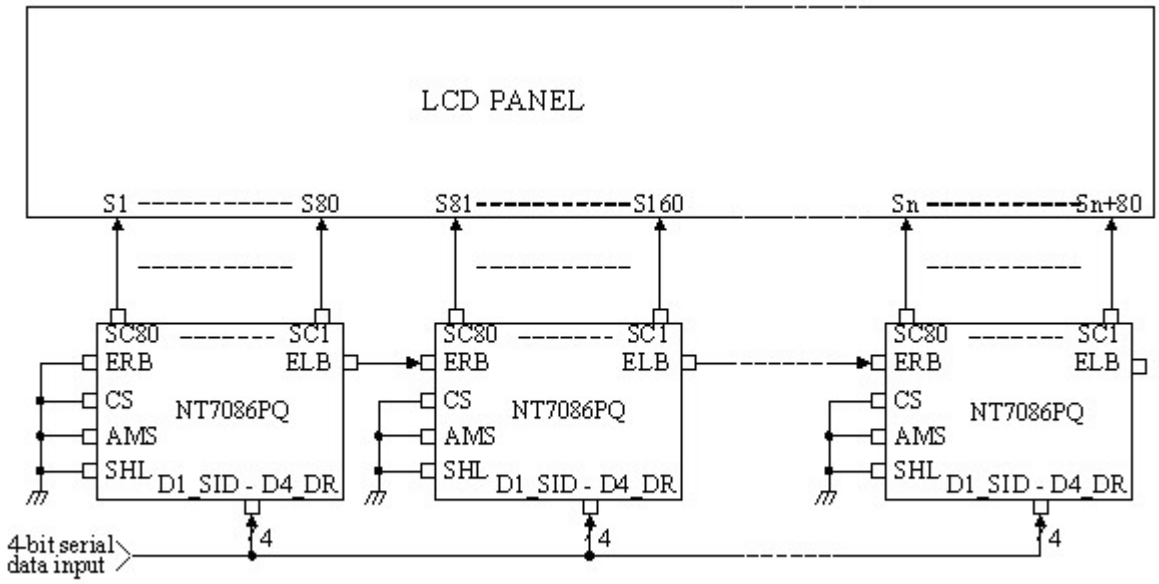
b) Upper view (SHL= H, AMS= H)



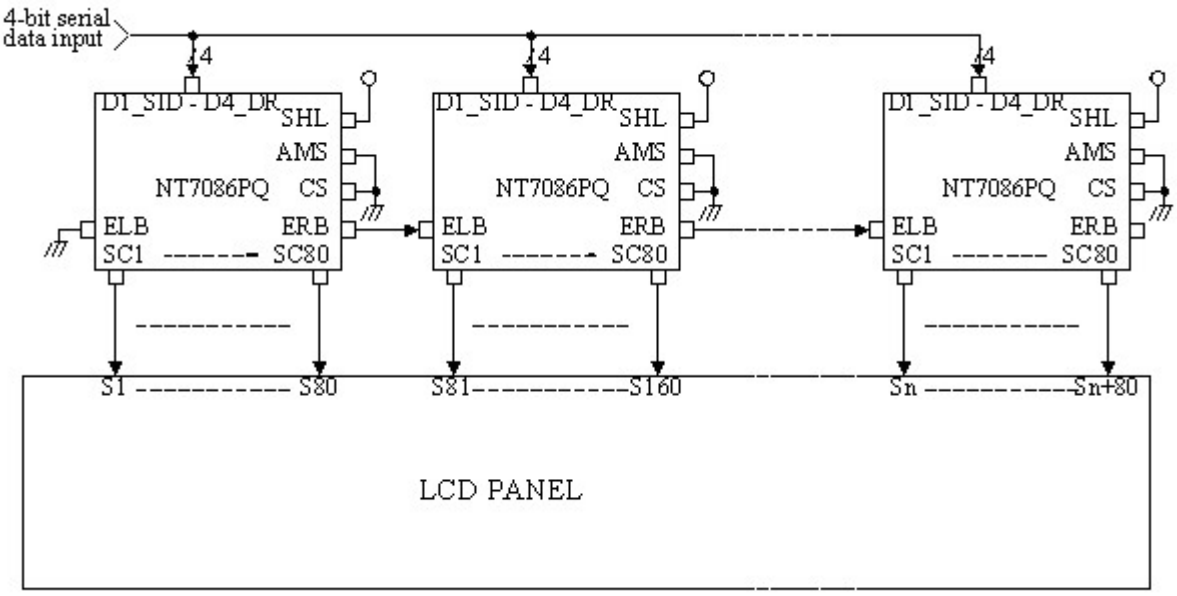
Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

4-bit parallel interface mode (80 Ch. Segment driver)

a) Lower view (SHL= L, AMS = L)

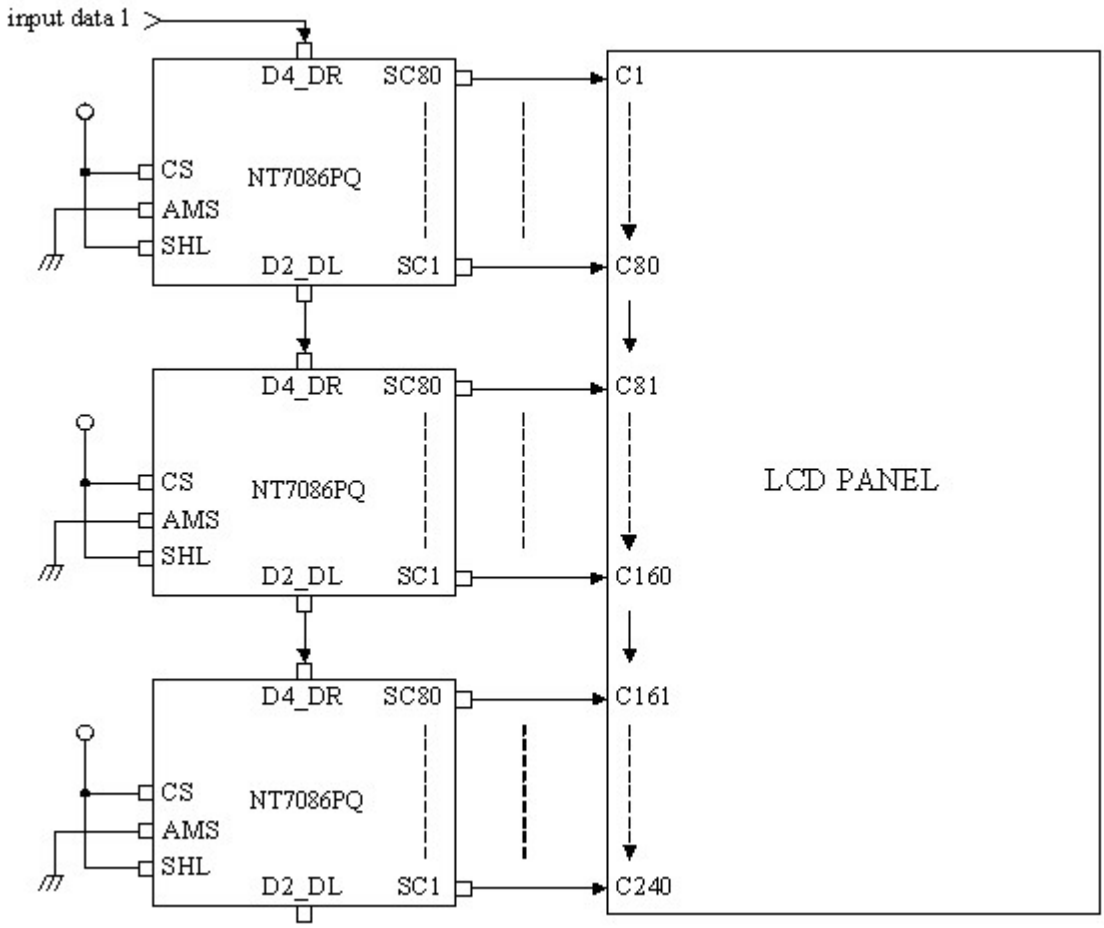


b) Upper view (SHL= H, AMS = L)



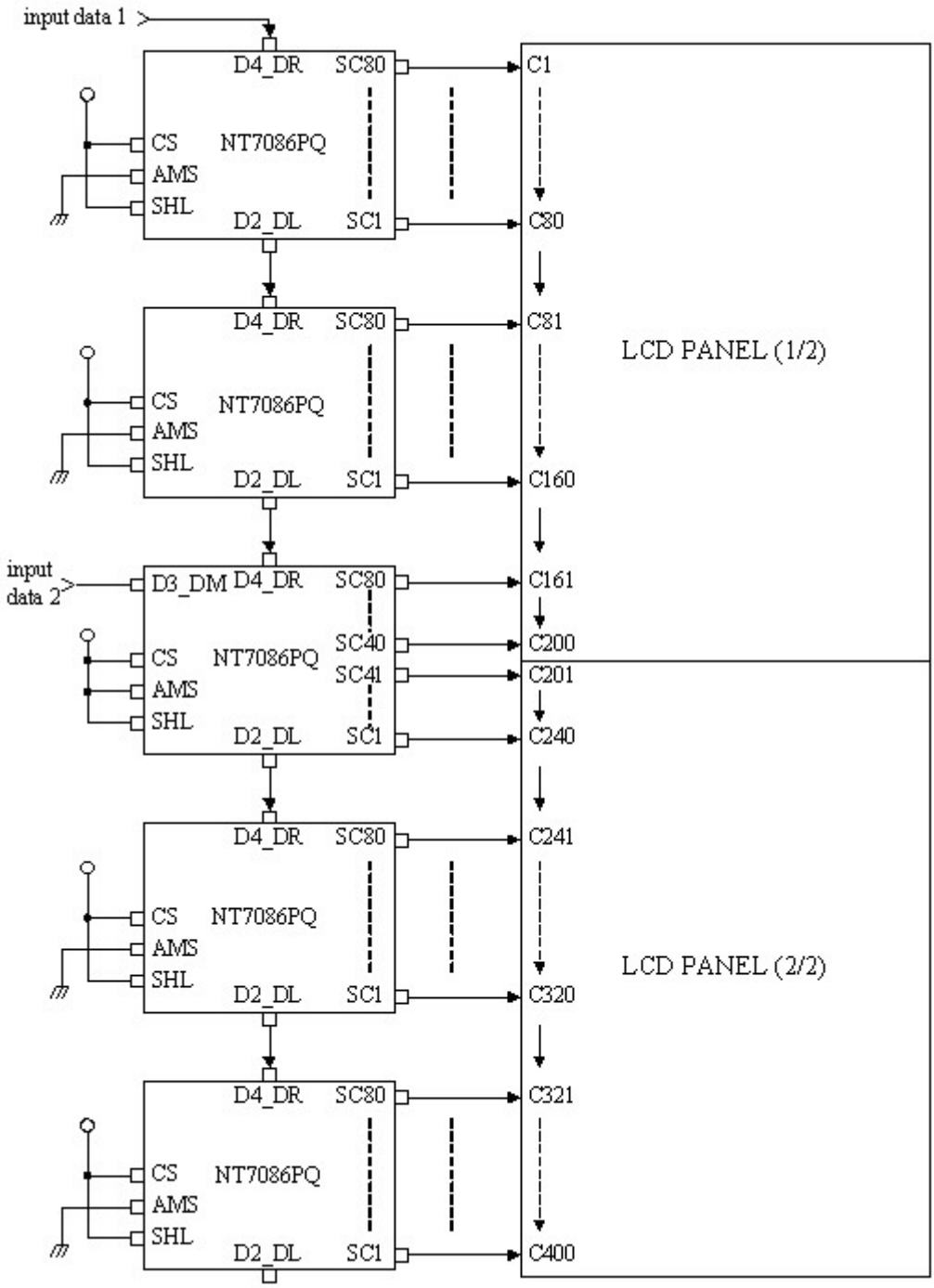
Crystalfontz This controller datasheet was downloaded from http://www.crystalfontz.com/controllers/

Single type interface mode (80 Ch. Common driver)



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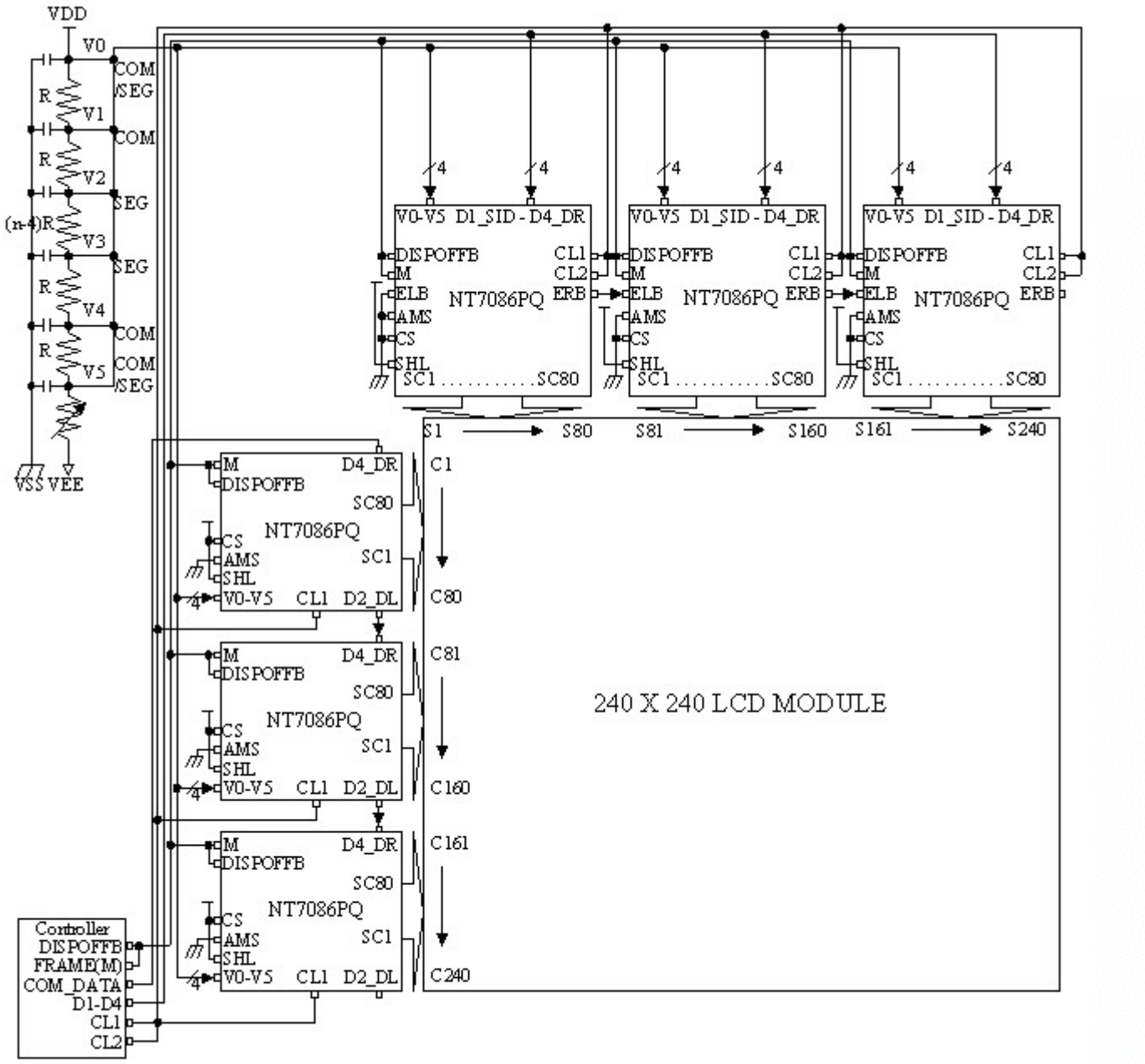
Dual-type interface mode (40 Ch. + 40Ch. Common driver)



NOTE: Using this application mode (dual-type common mode), the duty ratio can be reduced to half. In case, 1/200 duty can be used to driver the 400 common LCD panel.

Crystalfontz This controller datasheet was downloaded from **http://www.crystalfontz.com/controllers/**

APPLICATION CIRCUIT EXAMPLE



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PRECAUTIONS

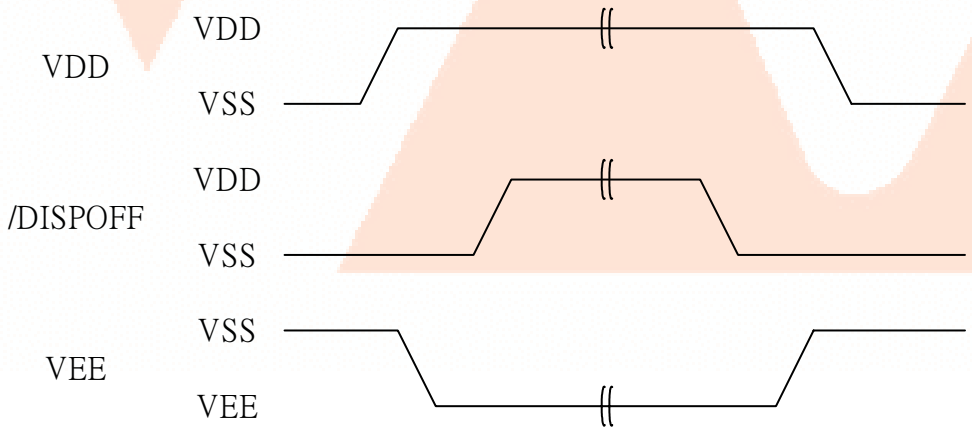
Precautions when connecting or disconnecting the power supply

This IC has a high-voltage LCD driver, so it may be permanently damaged by a high current which may flow if voltage is supplied to the LCD drive power supply while the logic system power supply is floating. The details are as follows.

When connecting the power supply, connect the LCD drive power after connecting the logic system power. Furthermore, when disconnecting the power, disconnect the logic system power after disconnecting the LCD drive power.

And when connecting the logic power supply, the logic condition of this IC inside is insecure. Therefore connect the LCD drive power supply after resetting logic condition of this IC inside on /DISPOFF function. After that, cancel the /DISPOFF function after the LCD drive power supply has become stable. Furthermore, when disconnecting the power, set the LCD drive output pins to level V5 on /DISPOFF function. Then, disconnect the logic system power after disconnecting the LCD drive power.

When connecting the power supply, follow the recommended sequence shown here.



溫度監控系統V2.52a

新德科技　　無鉛製程　IR Reflow Profile

